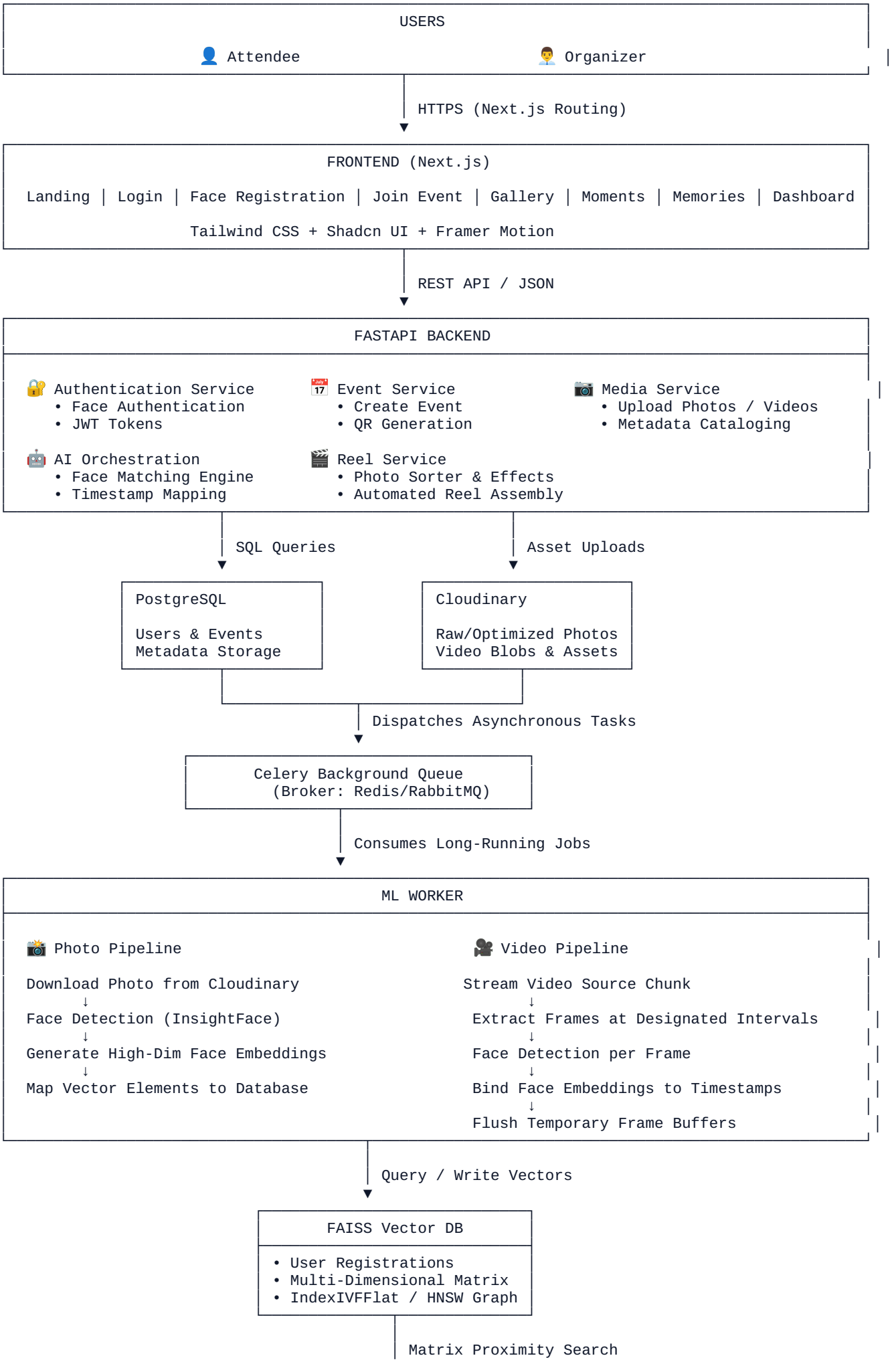
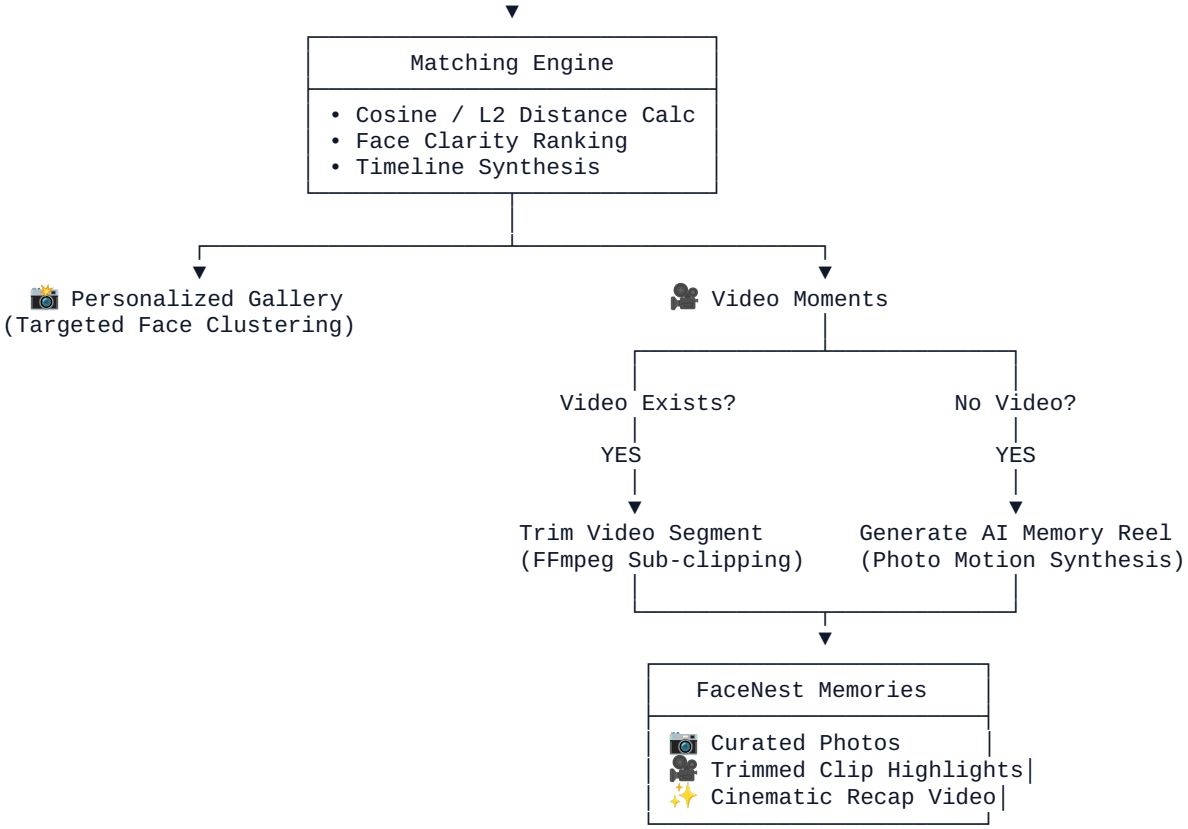

USERS
Attendee
Organizer
HTTPS (Next.js Routing)
FRONTEND (Next.js)
Landing | Login | Face Registration | Join Event | Gallery | Moments | Memories | Dashboard
Tailwind CSS + Shadcn UI + Framer Motion
REST API / JSON
FASTAPI BACKEND
Authentication Service
• Face Authentication
• JWT Tokens
Event Service
• Create Event
• QR Generation
Media Service
• Upload Photos / Videos
• Metadata Cataloging
AI Orchestration
• Face Matching Engine
• Timestamp Mapping
Reel Service
• Photo Sorter & Effects
• Automated Reel Assembly
SQL Queries
Asset Uploads
PostgreSQL
Users & Events
Metadata Storage
Cloudinary
Raw/Optimized Photos
Video Blobs & Assets
Dispatches Asynchronous Tasks
Celery Background Queue
(Broker: Redis/RabbitMQ)
Consumes Long-Running Jobs
ML WORKER
Photo Pipeline
Download Photo from Cloudinary
Face Detection (InsightFace)
Generate High-Dim Face Embeddings
Map Vector Elements to Database
Video Pipeline
Stream Video Source Chunk
Extract Frames at Designated Intervals
Face Detection per Frame
Bind Face Embeddings to Timestamps
Flush Temporary Frame Buffers
Query / Write Vectors
FAISS Vector DB
• User Registrations
• Multi-Dimensional Matrix
• IndexIVFFlat / HNSW Graph
Matrix Proximity Search


Matching Engine
• Cosine / L2 Distance Calc
• Face Clarity Ranking
• Timeline Synthesis
📸 Personalized Gallery
(Targeted Face Clustering)
🎥 Video Moments
Video Exists?
No Video?
YES
YES
Trim Video Segment
(FFmpeg Sub-clipping)
Generate AI Memory Reel
(Photo Motion Synthesis)
FaceNest Memories
📸 Curated Photos
🎥 Trimmed Clip Highlights
✨ Cinematic Recap Video